L1 · Introduction & Probability recap

- **IT = art of redundancy:** remove it when it hurts (compression/source coding), add it when it helps (channel coding). Info & statistics = two sides of one coin.
- **Repetition R3 over BSC(f):** decode = majority vote (Bayes: $\hat{s}=\text{argmax}_s\, P(r|s)P(s)$). $p_b = 3f^2(1-f)+f^3 \approx 3f^2$: error $O(f)\to O(f^2)$, but rate ÷3. Shannon: error-free comms at non-zero rate possible (rate < C).
- **Source code goals:** short + easily decodable; frequent symbols → shorter words.
- **Logs:** $\log_b(xy)=\log x+\log y$; $\log x^n=n\log x$; $\log_b x=\log_a x/\log_a b$; $\log 1=0$; $\log(x+y)$ does NOT simplify. $\log\equiv\log_2$; $\ln 2\approx 0.69$.
- **Convex-down ($\cup$, $f(\lambda x_1+(1-\lambda)x_2) \le \lambda f(x_1)+(1-\lambda)f(x_2)$):** prove via $f''\ge 0$, affine maps $f(Ax+b)$, non-neg. weighted sums. log and $-x\log x$ are concave ($\cap$).
- **Jensen:** f convex-down ⇒ $E[f(x)] \ge f(E[x])$ (reversed for concave, e.g. $E[\log x] \le \log E[x]$).
- **RV** $X:\Omega\to E$ measurable fn; $P(\omega)\ge 0$, $P(\Omega)=1$. Discrete: $\Sigma p=1$; continuous: $\int p=1$ (density may exceed 1, $\int_a^b p \le 1$). $E_p[f]=\Sigma p(x)f(x)$; $\text{Var}=E[(X-\mu)^2]=E[X^2]-\mu^2$. i.i.d. = independent + identically distributed.
- **Gaussian:** $N(\mu,\Sigma)$: $p=(2\pi)^{-d/2}|\Sigma|^{-1/2}\exp(-\tfrac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu))$. 68.3/95.5/99.7% within 1/2/3σ.
- **Limits:** (ε,δ) def; $\lim_{x\to 0+} \ln x=-\infty$.
- **Limit defs:** $\lim_{x\to p} f=L$ iff $\forall\varepsilon\,\exists\delta$: $0<|x-p|<\delta \Rightarrow |f-L|<\varepsilon$. Infinite limit: $\forall N\,\exists\delta \Rightarrow f>N$. At infinity: $\forall\varepsilon\,\exists c$: $x>c \Rightarrow |f-L|<\varepsilon$ (e.g. $1/x^2\to 0$).
- **Sets/functions:** $|S|$ cardinality, $\subset$ subset, S^c complement. Injective: $f(x_1)=f(x_2)\Rightarrow x_1=x_2$; bijective = injective + onto ⇏ invertible.
- **RV ingredients:** sample space Ω (all outcomes), event space E (measurements), P. Ex: 3 coins, X=#heads, $\Omega=\{H,T\}^3$, $E=\{0,1,2,3\}$. Notation: X RV, x realisation, $\mathcal{X}$ alphabet, $X\sim p(x)$, $p_i=p(x_i)$.
- **Source-coding example:** one-hot = 3 bit/pixel; variable length (Blue→0, Red→10, White→11) = 1.56 bit/pixel. BSC: $P(y=0|x=0)=P(y=1|x=1)=1-f$; $P(\text{flip})=f$. Beyond Shannon: multi-variable interactions $H(Z|X,Y)$, $I(X;Y;Z)$.
- **Course:** Part I source coding · II channel coding · III stats & advanced. Assessment 20% coursework, 80% exam.

L2 · Entropy & source coding theorem

- **Surprisal** $h(x)=\log(1/p(x))$. **Entropy** $H(X)=\Sigma p(x)\log(1/p(x))=E[h(X)]$. bits ($\log_2$) / nats (ln). $0\log 0=0$.
- **Binary entropy** $H_2(p)=-p\log p-(1-p)\log(1-p)$; max 1 bit at p=½. ($p=0.1\to H_2=0.47$; $h(0.2)=2.3$ bit; $h(0.1)=3.3$ bit.)
- **Bounds:** $0 \le H(X) \le \log|X|$. Min: Kronecker delta; max: uniform.
- **Independent:** $h(x,y)=h(x)+h(y)$; $H(X,Y)=H(X)+H(Y)$ (additive).
- **Worked (mock Q1):** $p=\{.25,.25,.2,.2,.1\}$:
  $H=2(.25\cdot 2)+2(.2\cdot\log 5)+.1\cdot\log 10=1+0.929+0.332=$ **2.261 bit**. Lengths $\lceil -\log p\rceil=\{2,2,3,3,4\}$; Kraft $\Sigma 2^{-\ell}=.25+.25+.125+.125+.0625=0.8125\le 1$ ⇒ a prefix code with these lengths exists (Kraft ≤1 ⇔ lengths are realisable by a prefix code). $L=1+1.2+.4=2.6$ bit ≥ $H=2.261$ (SCT/converse: L≥H for any code; Huffman is optimal).
- **Shannon axioms:** $i(1)=0$; $di/dp \le 0$; $i(pq)=i(p)+i(q) \Rightarrow i(p)=-\log_b p$ (unique).
- **Guessing games:** uniform ⇒ $h(x)$=#yes/no questions (63-game: 6 bits). Submarine: total info always log 64=6 bit regardless of when hit. Info = probability-mass exclusion (Hartley). Design experiments with evenly probable outcomes.
- **Differential entropy** $H(X)=-\int f\log f\,dx$. Discretising: $H(X^\Delta)=-\Sigma\,\Delta f(x_i)\log f(x_i) - \log\Delta \to$ diverges; drop $-\log\Delta$. Can be negative; U[0,a]: $H=\log a$; 2^H=volume.
- **Gaussian:** surprisal $h(x)=\ln(\sigma\sqrt{2\pi})+\tfrac{1}{2}((x-\mu)/\sigma)^2 = \ln\sigma + \text{MSE}/2\sigma^2 + C$. **$H=\tfrac{1}{2}\ln|2\pi e\Sigma|$** (1D: $\tfrac{1}{2}\ln(2\pi e\sigma^2)=\tfrac{1}{2}(1+\ln 2\pi\sigma^2)$). D-dim: derived via $E[(x-\mu)^T\Sigma^{-1}(x-\mu)]=\text{tr}(\Sigma^{-1}\Sigma)=D$.
- **Meanings of H:** avg surprisal $E[h(X)]$; min expected #yes/no questions (exact for uniform, asymptotic otherwise); min compressed length per symbol; generalised variance (how quickly p fills the space; Gaussian: grows with $\ln|\Sigma|$ / avg MSE); $H(X)=\log|X|-D(p\|u)$. Continuous: only differential H (not invariant, can be <0).
- **Code:** $C:X\to D^*$; $\ell(x)$ length; $L=E[\ell(x)]$. Binary $|D|=2$, d-ary $|D|=d$. Uniform: $L=\log|X|$ (raw bit content); if $|X|$ not power of 2 use $\lceil\log|X|\rceil$ ("extra bit"). Ignoring 2% rare symbols saved 6→5 bits (lossy).
- **wLLN:** $\bar{Y}_n=(1/n)\Sigma X_i$: $E=\mu$, $\text{Var}=\sigma^2/n$; $\bar{Y}_n \to^P \mu$: $\lim \Pr(|\bar{Y}-\mu|<\varepsilon)=1$. High-dim Gaussian: points lie in a thin shell, $\text{Std}(r^2)/E[r^2]=O(1/\sqrt{n})\to 0$.
- **AEP:** $-(1/n)\log p(x_1\ldots x_n) \to H(X)$ (via wLLN on $-(1/n)\Sigma\log p(x_i)$). All typical samples ~equally probable.
- **Typical set** $A_\varepsilon^{(n)}$: $2^{-n(H+\varepsilon)} \le p(x^n) \le 2^{-n(H-\varepsilon)}$. **Props:** (1) $x^n\in A \Rightarrow |-(1/n)\log p(x^n) - H|\le\varepsilon$, $p\approx 2^{-nH}$; (2) $P(x^n\in A)>1-\varepsilon$; (3) $|A| \le 2^{n(H+\varepsilon)}$ [$1=\Sigma p \ge \Sigma_A p \ge |A|2^{-n(H+\varepsilon)}$]; (4) $|A| \ge (1-\varepsilon)2^{n(H-\varepsilon)}$.
- **Coin intuition (bias p):** $p^r(1-p)^{n-r}$ favours few 1s, C(n,r) favours n/2 ⇒ balance at r/n=p. Only $\approx 2^{nH}$ of $|X|^n$ strings happen. Entropy = log(#events that almost always happen)/log(#possible) = fraction of volume; Gaussian $H \propto \log V + C$.
- **Source Coding Thm (SCT):** X^n i.i.d.~p; $\forall\varepsilon>0$, large $n \Rightarrow \exists$ invertible code with $E[(1/n)\ell(X^n)] \le H(X)+\varepsilon$. *Converse:* no invertible code has $L<H(X)$. Achievability proof: flag bit 0 + index in A ($\ell\le 1+\lceil\log|A|\rceil\le n(H+\varepsilon)+2$) else flag bit 1 + index in X^n ($\ell\le n\log|X|+2$). ⇒ $L\le n(H+\varepsilon')$, $\varepsilon'=\varepsilon+\varepsilon\log|X|+2/n$. Converse: subset $C\subset A$ with $|C| \approx 2^{n(H-\kappa)}$ has $P(C)\approx 2^{-n\kappa}\to 0$.

L3 · Symbol codes, KL, MLE

- **No free lunch:** cannot shorten all strings: lossy (collisions) or lossless (some strings longer).
- **Code types (nested):** nonsingular ($x\ne x' \Rightarrow C(x)\ne C(x')$) ⊃ uniquely decodable (extension $C^*(x_1\ldots x_n)=C(x_1)\ldots C(x_n)$ nonsingular) ⊃ **prefix** (instantaneous; no codeword is a prefix of another; leaves of a tree). Morse: nonsingular, not UD. Invertible ≠ easy to invert (RSA).
- **Kraft:** prefix code lengths satisfy $\Sigma_i D^{-\ell_i} \le 1$; conversely lengths satisfying it ⇒ ∃ prefix code. (Also holds for UD codes.) Proof: $\le D^{\ell_{max}}$ leaves; word of length ℓ_i kills $D^{\ell_{max}-\ell_i}$ leaves. "Codeword supermarket" budget.
- **Optimal lengths** (Lagrange $L=\Sigma p_i\ell_i+\lambda(\Sigma D^{-\ell_i}-1)$; $\partial L/\partial\ell_i=p_i-\lambda D^{-\ell_i}\ln D=0$, $\lambda=1/\ln D$): **$\ell_i^*=-\log_D p_i$, $L^*=H_D(X)$**; $\partial^2 L=D^{-\ell}\ln D\ge 0$ ⇒ global min. Entropy = min length of any symbol code. Duality: lengths ℓ ⇒ implicit probs $p\propto D^{-\ell}$. Non-integer ⇒ round up $\ell=\lceil -\log p\rceil$ (still satisfies Kraft).
- **Lagrange:** min f(x) s.t. g(x)=0: $L=f+\lambda\cdot g$, solve $\nabla_{x,\lambda}L=0$ (k+c equations).
- **Huffman:** repeatedly merge the two least probable symbols into a tree node until one remains; read codes root→leaf. Optimal: $L(C^*)\le L(C')$ for any UD code. Ex: $p=\{.25,.25,.2,.15,.15\}\to\ell=\{2,2,2,3,3\}$.
- **Cross-entropy** $H(p,q)=-\Sigma p\log q$ = expected length of q-optimal code on p-samples. **KL** $D(p\|q)=\Sigma p\log(p/q)=H(p,q)-H(p)$ = extra cost of wrong code.
- **Chain of reasoning (exam):** $\ell^*=-\log p \Rightarrow L^*=H$; wrong q ⇒ $L=H(p,q)=H(p)+D(p\|q)$; $L\ge H$ follows from Gibbs ($D\ge 0$). ML on data ⇔ min $D(p\|q)$ ⇔ min code length.
- **Gibbs:** $D(p\|q)\ge 0$, =0 iff p=q. Proof: $-D=\Sigma p\log(q/p) \le \log\Sigma q = 0$ (Jensen, log concave). KL convex in q; asymmetric (not distance); =∞ if ∃x: q=0, p>0. $D(p\|u)=\log|X|-H(p) \Rightarrow H\le\log|X|$. Ex: $D(p_2\|p_3)=\infty$.
- **Continuous KL** = Σ → ∫f log(f/g): log Δ cancels ⇒ ≥0.
- **Wrong-code ex:** code for $q=\{\tfrac{1}{2},\tfrac{1}{4},\tfrac{1}{8},\tfrac{1}{8}\}$ (C={0,10,110,111}) costs 1.75 bit on q but 2.66 bit on $p=\{0,\tfrac{1}{3},\tfrac{1}{3},\tfrac{1}{3}\}$. KL examples (a–d): $D(p_1\|p_2)=0.25$, $D(p_2\|p_3)=\infty$, $D(p_3\|p_2)=0.5$, $D(p_3\|p_1)=1$ bit.
- **Prefix tree facts:** complete D-ary tree has $D^{\ell_{max}}$ leaves; adding a codeword removes all its descendants. Huffman ex: a→00 b→10 c→11 d→010 e→011.
- **SCT summary (symbol codes):** $\forall p(x)\ \exists$ prefix code with L close to H (achievability), none with L<H (converse). C&T proves SCT via lossless block code; MacKay via lossy (error only outside A).
- **KL of Gaussians:** $D(p\|q)=\tfrac{1}{2}[\sigma_p^2/\sigma_q^2 - 1 - \ln(\sigma_p^2/\sigma_q^2) + (\mu_p-\mu_q)^2/\sigma_q^2]$; $H(p,q)=\tfrac{1}{2}\ln 2\pi\sigma_q^2+(\sigma_p^2+(\mu_p-\mu_q)^2)/2\sigma_q^2$. Narrow Gaussians: KL grows faster with Δμ.
- **MLE:** $\theta^*=\text{argmax}\, p(x;\theta)$; i.i.d.: $L(\theta)=(1/n)\Sigma\ln p(x_i;\theta) \to E_p[\ln p(x;\theta)]$. Gaussian mean: $\mu^*=(1/n)\Sigma x_i$.
- **Good compressor = good predictor:** $q(x)=2^{-\ell(x)}/z$, $z=\Sigma 2^{-\ell}\le 1$ (Kraft number). $E_p[\ell-\ell^*]=\log(1/z)+D(p\|q) \ge 0$; =0 iff Kraft equality & p=q.

L4 · Joint entropy & mutual information

- **Joint probs:** marginal $p(x)=\Sigma_y p(x,y)$; cond. $p(x|y)=p(x,y)/p(y)$; product $p(x,y)=p(x|y)p(y)$; sum $p(x)=\Sigma p(x|y)p(y)$; Bayes $p(x|y)=p(y|x)p(x)/p(y)$.
- **Independence:** $X\perp Y \Leftrightarrow p(x,y)=p(x)p(y)$ ($\Leftrightarrow p(x|Y)=p(x)$). Cond.: $X\perp Y|Z \Leftrightarrow p(x,y|z)=p(x|z)p(y|z)$. Conditioning can create/destroy independence: Y=f(X),Z=g(Y): $X\perp Z$ but $X\perp Z|Y$; X,Y fair, Z=X+Y mod 2: $X\perp Z$ but $X\perp Z|Y$.
- **Bayesian net:** $p(X)=\Pi_k p(X_k|pa_k)$. **Markov chain** X–Y–Z: $p(X_t|X_{t-1}\ldots X_1)=p(X_t|X_{t-1})$ (order p: last p). X–Y–Z ⇒ $X\perp Z|Y$.
- **Joint** $H(X,Y)=-\Sigma p(x,y)\log p(x,y)$ (symmetric). **Cond.** $H(X|Y)=\Sigma_y p(y)H(X|Y=y)$. Props: $0 \le H(X|Y) \le \log|X|$; =0 if X=f(Y); $X\perp Y \Rightarrow H(X|Y)=H(X)$.
- **Cond. KL:** $D(p(X|Y)\|q(X|Y))=\Sigma_y p(y)D(p(X|y)\|q(X|y)) \ge 0$.
- **Chain rule: $H(X,Y)=H(X|Y)+H(Y)=H(Y|X)+H(X)$** (proof: $p(x,y)=p(x|y)p(y)$, split log). Ex: $H(X,Y)=2.25$, $H(X)=1$, $H(Y)=1.41$, $H(X|Y)=0.84$, $H(Y|X)=1.25$.
- **Isomorphism** (bijective f): $H(X)=H(Y)$ in discrete (via both chain rules, $H(X|Y)=H(Y|X)=0$); $I(X;Y)$ invariant even in continuous.
- **Stationary process:** $p(X_1..X_t)=p(X_{1+s}..X_{t+s})$. **Entropy rate** $h_1=\lim(1/n)H(X_1..X_n)$ = **innovation** $h_2=\lim H(X_n|X_1..X_{n-1})$ (stationary). More conditioning ⇒ less entropy; $H(X_n|X_{n-t}..X_{n-1})$ decreases to h at t=Markov order. Coin-in-box (flip w.p. p): $h=H_2(p)$. Stream codes: Lempel-Ziv (zip).
- **Mutual information** $I(X;Y)=\Sigma p(x,y)\log[p(x,y)/(p(x)p(y))]$ = **$H(X)+H(Y)-H(X,Y) = H(X)-H(X|Y) = H(Y)-H(Y|X) = D(p(X,Y)\|p(X)p(Y))$**. Symmetric; ≥0; =0 iff $X\perp Y$; $I \le \min\{H(X),H(Y)\}$ (discrete). Notation: $I(X,Y;Z)$ joint vs ; separator.
- **Information doesn't hurt:** $H(X|Y) \le H(X)$ (avg; $H(X|Y=y)$ can exceed $H(X)$).
- **Joint-table method:** get p(x),p(y) by summing rows/cols; $H(X|Y)=\Sigma_y p(y)H(X|y)$; check $H(X|Y)=H(X,Y)-H(Y)$. Die ex: X=even, Y=>3 ⇒ joint = intersections of preimages $p(X=x,Y=y)=p(\omega\in X^{-1}(x)\cap Y^{-1}(y))$.
- **Inequalities cheat-list:** $H(X,Y) \le H(X)+H(Y)$ (eq. iff $\perp$); $H(X,Y) \ge \max(H(X),H(Y))$; $H(X|Y)\ge 0$; $I(X;Y)\ge 0$; $I(X;Z)\le I(X;Y)$ if X–Y–Z; $I(X;Y)\le H(X)$ (discrete); for continuous H can be <0 but I stays ≥0.
- **CMI** $I(X;Y|Z)=H(X|Z)+H(Y|Z)-H(X,Y|Z)=H(X|Z)-H(X|Y,Z)=D(p(X,Y|Z)\|p(X|Z)p(Y|Z)) \ge 0$; =0 iff $X\perp Y|Z$.
- **MI chain rule: $I(X,Y;Z)=I(X;Z)+I(Y;Z|X)$**.
- **Data processing ineq.:** X–Y–Z ⇒ **$I(X;Z) \le I(X;Y)$**
  ($I(X,Y,Z)=I(X;Z)+I(X;Y|Z)=I(X;Y)+I(X;Z|Y)$, last =0).
- **MI properties summary:** $I(X;Y)=E[\log p(x,y)/p(x)p(y)]$; $I(X;X)=H(X)$; $I(X;f(Y)) \le I(X;Y)$ (DPI); reduction of uncertainty in X after knowing Y; Gaussian I depends only on correlation ρ; zero ρ ⇏ I=0 in general.
- **Venn:** $H(X)\leftrightarrow\mu(A)$, $H(X,Y)\leftrightarrow\mu(A\cup B)$, $H(X|Y)\leftrightarrow\mu(A\setminus B)$, $I\leftrightarrow\mu(A\cap B)$. Co-information $I(X;Y;Z)=H(XYZ)-H(XY)-H(YZ)-H(XZ)+H(X)+H(Y)+H(Z)$ can be negative (signed measure).
- **Gaussian MI: $I(X;Y)=-\tfrac{1}{2}\ln(1-\rho^2)$**=$\tfrac{1}{2}\ln(\sigma_{yy}/\sigma^2_{y|x})$ (explained variance). ρ=0 ⇒ I=0 only for Gaussians; uncorrelated ≠ independent in general; MI = generalised dependence.
- **MI & prediction (Barber-Agakov):** $I(X;Y)=H(Y)+E[\log p(y|x)] \ge H(Y)+E_{p(x,y)}[\log q_\theta(y|x)]$ (Gibbs). Equality if $p(y|x)\in Q$; MI upper-bounds best predictor.

L5 · Channel coding

- **DMC:** input alphabet X, output Y, p(y|x); extended $p(y^n|x^n)=\Pi p(y_i|x_i)$. **BSC(f):** flips w.p. f; uniform input: **$I=1-H_2(f)$** ($H(Y|X)=H_2(f)$, $H(Y)=1$).
- **Capacity** $C=\max_{p(x)} I(X;Y)$; maximiser p*(x) = capacity-achieving dist. Interpretation: max rate (bits/use) at which info can be sent with vanishing error. Recipe: param. p(X=1)=α, write I(α)=H(Y)−H(Y|X), set dI/dα=0.
- **Noisy typewriter** (|X|=|Y|=27, letter or 2 neighbours w.p. ⅓): H(Y)=log27, H(Y|X)=log3, I=log9 for uniform X; use every 3rd letter ⇒ error-free, C=log9. Pick non-confusable inputs (disjoint output sets). Counting: $2^{nH(Y)}/2^{nH(Y|X)}=2^{nI(X;Y)}$ distinguishable inputs.
- **(n,K) block code:** 2^K codewords length n; rate **R=K/n** bits/channel use. Decoder $\hat{s}=Dec(y^n)$ (+failure symbol). Mean block error $p_B=\Sigma p(s)p(\hat{s}\neq s|s)$; maximal $p_{BM}=\max_s p(\hat{s}\neq s|s)$; bit error p_b.
- **Noisy channel coding thm:** (1) ∀ε>0, R<C, large n ⇒ ∃ code of rate R with $p_{BM}<\varepsilon$. (2) With p_b acceptable, rates up to **$R(p_b)=C/(1-H_2(p_b))$** achievable. (3) Rates above $R(p_b)$ not achievable.
- **Jointly typical set** $J_\varepsilon^{(n)}$: $|-(1/n)\log p(x^n)-H(X)|<\varepsilon$, same for y^n, and $|-(1/n)\log p(x^n,y^n)-H(X,Y)|<\varepsilon$. Props: (1) $p(x^n,y^n)\approx 2^{-nH(X,Y)}$; (2) $Pr(J)\to 1$ (union bound on complements); (3) $|J|\le 2^{n(H(X,Y)+\varepsilon)}$; (4) independent $\tilde{x}^n,\tilde{y}^n\sim p(x^n)p(y^n)$: $Pr(J)\le$ **$2^{-n(I(X;Y)-3\varepsilon)}$**.
- **Proof ingredients:** (a) random codes (2^{nR} words $\sim p(x^n)$; average $\langle p_B\rangle$ easy; "weighing babies": avg ≤δ ⇒ some code ≤δ); (b) typical-set decoder (success iff exactly one s with $(x^n(s),y^n)$ jointly typical); (c) errors: $x^n(s)$ not typical $\to 0$ (Prop 2), or other s′ typical $\le (2^{nR}-1)2^{-n(I-3\varepsilon)}\le 2^{-n(I-R-3\varepsilon)}\to 0$ iff $R<I-3\varepsilon$ (use $p^*\Rightarrow R<C-3\varepsilon$); union bound; (d) **expurgation:** drop worst half of words ⇒ $p_{BM}\le 2p_B$, rate $R'=R-1/n$.
- **Repetition R_N:** R2 detects 1 flip, R3 corrects 1 flip. **Parity:** XOR = sum mod 2.
- **Worked: Z channel** p(Y=0|X=0)=1, p(Y=0|X=1)=½. Uniform X: p(Y=0)=¾ ⇒ $H(Y)=H_2(¼)=0.811$; $H(Y|X)=½\cdot 0+½\cdot 1=0.5$ ⇒ **I=0.311 bit**. General $\alpha=p(X=1)$: $p(Y=1)=\alpha/2$, $I(\alpha)=H_2(\alpha/2)-\alpha$; $dI/d\alpha=½\log[(1-\alpha/2)/(\alpha/2)]-1=0\Rightarrow(1-\alpha/2)=2\alpha\Rightarrow$ **α*=0.4**, $C=H_2(0.2)-0.4=\log(5/4)=0.322$ bit. BSC: $C=1-H_2(f)$ at uniform input.
- **Bit-error tradeoff:** allowing p_b raises max rate to $C/(1-H_2(p_b))$; $p_b\to 0$ recovers C. Three regions: (1) any p_b incl. 0 (R<C); (2) faster codes with $p_b>0$; (3) prohibited. R3 over BSC: rate ⅓, $p_b\approx 3f^2$.
- **Hamming encode ex:** $s=(d_1d_2d_3d_4)$; $t=(p_1p_2d_1p_3d_2d_3d_4)$. Decode: compute z=Hr; if z≠0 flip bit at position $(z_3z_2z_1)_2$ [z_1=LSB]; read data bits 3,5,6,7. Detects 2 errors but mis-corrects them.
- **Hamming(7,4):** $n=2^m-1=7$, K=4 data, m=3 parity; $2^4=16$ codewords; layout $p_1p_2d_1p_3d_2d_3d_4$ (parity at positions 1,2,4). $p_1=d_1+d_2+d_4$, $p_2=d_1+d_3+d_4$, $p_3=d_2+d_3+d_4$ (mod 2).
- **Matrices:** $H\in B^{m\times n}$, column j = binary of j (LSB top): H=[1010101; 0110011; 0001111]. $G\in B^{K\times n}$: data columns = identity I_K, parity columns = data rows of H^T. **$t=G^Ts$ mod 2; $r=t+n$ mod 2; syndrome $z=Hr$ mod 2=Hn** (Ht=0). Syndrome = binary index of flipped bit (e.g. 110→bit 3). Corrects any single flip. Systematic = data bits first, parity last.
- **Hamming distance** d=#differing bits; min distance t ⇒ corrects $\lfloor(t-1)/2\rfloor$ errors (Hamming(7,4): t=3). Sphere packing; "perfect" = spheres fully packed.
- **Source-channel separation:** optimal error-free rate for V through p(y|x) achieved by separate source code (compress to H(V)) + channel code. Summary: **source coding: H = optimal compression; channel coding: I = optimal transmission.**

L6 · Probabilistic inference

- **Descriptive vs inferential** statistics (param. estimation, hypothesis testing). Frequentist: true fixed θ, repeated experiments. Bayesian: beliefs about θ (θ is an RV).
- **NHST:** H_0 (null, "business as usual") vs H_1. Rejection region R: x∈R ⇒ reject H_0. Can never "accept" H_0. **Simple**: point $\theta=\theta_0$; **composite**: region $\theta\in\Theta_0$; **nested**: H_0 fixes a parameter of H_1 (e.g. a=0 in y=ax+b).
- **Type I** α=P(x∈R|H_0) (false positive; boy cried wolf). **Type II** β=P(x∉R|H_1) (false negative). **Power**=1−β. Bound α, maximise power.
- **Neyman-Pearson:** for simple $H_0:\theta_0$ vs $H_1:\theta_1$, most powerful test of size α is R={x: $p_{\theta_1}(x)/p_{\theta_0}(x) > k$} with $P_{\theta_0}(R)=\alpha$. Proof: compare R vs R′ equal α; R\R′ has ratio>k, R′\R ratio<k, equal null prob ϕ ⇒ $P_{\theta_1}(R)>P_{\theta_1}(R')$.
- **LRT:** $\Lambda=\sup_{\theta\in\Theta_1}p_\theta(x)/\sup_{\theta\in\Theta_0}p_\theta(x)$; optimal for point (N-P) & monotone-LR composite (Karlin-Rubin); not always optimal. i.i.d.: $\log\Lambda=\Sigma[\log p_{\theta_1^*}(x_i)-\log p_{\theta_0^*}(x_i)]\to nE[...]$; θ_j^* = MLE under each hypothesis. log Λ = frequentist evidence for H_1.
- **MI as LR:** H_0: p=f(x)g(y) vs H_1: general. $\sup\log p_{H_1}=-nH(X,Y)$, $\sup\log p_{H_0}=-n(H(X)+H(Y))\Rightarrow$ **I(X;Y)=(1/n)E[log Λ]**. Most info quantities are log-LRs.
- **Bayes:** $p(\theta|D)=p(D|\theta)p(\theta)/p(D)$: posterior=likelihood×prior/evidence. Predictive $p(D'|D)=\int p(D'|\theta)p(\theta|D)d\theta$. **MAP** θ=argmax $p(D|\theta)p(\theta)$. Evidence $p(D)=\int p(D|\theta)p(\theta)d\theta=E_{p(\theta)}[p(D|\theta)]$ ("replace sups with sums").
- **Gaussian posterior** (prior $\mu\sim N(\mu_0,\sigma_0^2)$, $x\sim N(\mu,\sigma^2)$): $\mu_1=(\sigma_0^2x+\sigma^2\mu_0)/(\sigma_0^2+\sigma^2)$, $1/\sigma_1^2=1/\sigma_0^2+1/\sigma^2$.
- **Bayesian hypothesis test:** $p(H|D)=p(D|H)p(H)/p(D)$, $p(D)=\Sigma p(D|H)p(H)$. Posterior odds = **Bayes factor $BF=p(D|H_1)/p(D|H_0)$** × prior odds. BF≫1 ⇒ H_1; ≈1 inconclusive; ≪1 ⇒ H_0 (can support either). Coin ex: H_0:q=½ ⇒ $p(D|H_0)=2^{-n}$; H_1:q~U(0,1) ⇒ $p(D|H_1)=n!(n-h)!/(n+1)!$; $BF=n!(n-h)!2^n/(n+1)!$.
- **Occam factor:** $p(D|H)\approx p(D|w_{MP},H)\times\sigma_{w|D}/\sigma_w$ = best-fit likelihood × Occam factor; small when large prior space but small fitting region. Complex models spread evidence thinly.
- **MDL (Shannon's razor):** L(D)=L(H_i)+L(w*|H_i)+L(D|w*,H_i); choose min; trade-off complexity vs fit.
- **Worked LLR (mock Q4):** H_0: exponential λ vs H_1: half-normal σ on x≥0 (both parametric families ⇒ **composite**, not nested (different families), so N-P does not directly apply: N-P guarantees optimal power only for simple vs simple; LR still "usually good"). MLEs: $\lambda^*=1/\bar{x}$, $\sigma^{*2}=\text{mean}(x^2)\equiv\bar{x}^2$. $\log p_{\lambda^*}=-n\ln\bar{x}-n$; $\log q_{\sigma^*}=(n/2)\ln(2/(\pi\bar{x}^2))-n/2$. **$\log\Lambda=n[\ln\bar{x}-½\ln(\bar{x}^2)+½\ln(2/\pi)+½]$** (bars: mean(x) vs mean(x^2)).
- **Coin vs die ex:** H_0 Bernoulli(½) vs H_1 Bernoulli(1/6); data HHTTTTHTTHTTTTHTTTT (5 heads/20): $p(D|H_0)=2^{-20}\approx 10^{-6}$, $p(D|H_1)=(1/6)^5(5/6)^{15}\approx 8\times 10^{-6}$ ⇒ LR≈8 favours die. If H_1 true: $\log p_{\theta_1}$ grows linearly in n.
- **Frequentist vs Bayes:** freq. uses sup (best θ); Bayes averages (evidence). Inference ≠ prediction: ML predicts, DS finds out. Posterior needs prior p(θ); D−θ−D′ is a Markov chain.

L7 · Estimating information

- Info quantities need p(x); only samples ⇒ **density estimation** then plug into formula. Types: 0 plug-in (discrete) · I parametric · II non-parametric · III variational.
- **Plug-in:** $\hat{p}(x_j)=\#(X=x_j)/n$, plug into $-\Sigma\hat{p}\log\hat{p}$. Binning continuous = usually bad.
- **Mock Q2 recipes:** (a) Gaussian estimator on X-shaped / non-linearly dependent data: Σ diagonal ⇒ $\hat{I}=0$ though dependent ⇒ unsuitable (model mismatch). (b) Non-param H(X): place kernel K_h on each x_i, $q(x)=(1/n)\Sigma K_h(x-x_i)$ (pick h by cross-validation), then $\hat{H}=-(1/n)\Sigma_i\log q(x_i)$ [or k-NN]. (c) $I=H(Y)-H(Y|X)\ge H(Y)+E_p[\log q(y|x)]$ since $E_p\log p(y|x)-E_p\log q(y|x)=D(p\|q)\ge 0$ (Gibbs); LLN: $E\approx(1/n)\Sigma\log q(y_i|x_i)$ for large n.
- **Parametric:** fit q_θ ($\theta^*=\text{argmax}\, q_\theta(x^n)$), $H=-E_{q_{\theta^*}}[\log q_{\theta^*}]$; Gaussian: $½\ln|2\pi e\Sigma|$. Garbage in, garbage out (Gaussian fit to non-linear dependence gives I=0).
- **Non-parametric:** $H=-E_p[\log q(x)]$; kernel $q(x)=(1/n)\Sigma K_h(x-x_i)$, $K_h=(2\pi h^2)^{-½}\exp(-(x-x_i)^2/2h^2)$; tune h via cross-validation. k-NN (Kraskov): adaptive width, area A with K nearest neighbours, locally uniform; more accurate.
- **Variational:** $H=\max_\theta\hat{H}_\theta(x^n)$ (estimation→optimisation). **Barber-Agakov:** $I(X;Y)\ge H(Y)+(1/n)\Sigma\log q_\theta(y_i|x_i)$ (any q; Gibbs). **Contrastive** (van den Oord): positives from p(x,y), negatives from p(x)p(y) (shuffle); classifier f_θ; $I\ge L_{CE}=-E\log[f(x_i,y_i)/\Sigma_{j=1}^M f(x_j,y_i)]$; optimal f=p(x,y)/(p(x)p(y)).
- **Bias**=$\langle\hat{I}\rangle-I$; **Variance**=$\langle(\hat{I}-\langle\hat{I}\rangle)^2\rangle$; both worsen with little data; cheap methods ⇒ more assumptions. Probabilistic numerics: estimation as inference.
- **Statistic** T(x); **sufficient** if θ−T−X Markov chain. R={x:T(x)>k}. **p-value**=P under H_0 of T ≥ observed; significant if p<α (0.05). Critical value: $\alpha=\int_{t_{crit}}^\infty p_{\theta_0}(T=t)dt$. Null distribution hard; **Wilks:** $2\log\Lambda\to\chi^2$ as $n\to\infty$ (finite n: simulate). Report p-values but not only p-values.
- **Surrogate data:** sample null empirically by randomising only the tested property. Independence: $\hat{I}(X;Y)$, shuffle $Y\to Y^S$, p = #{$\hat{I}(X;Y^S)\ge\hat{I}(X;Y)$}/#surrogates. Triplet: statistic/null/surrogate.
- **Power analysis** (sample size for effect): power=1−β=P(x∈R|H_1) depends on fixed $\theta\in\Theta_1$ & n. Simulate: pick n, μ>0; draw n samples from H_1; run NHST at α; repeat M times; power=fraction rejected. Increases with effect and n. Analytical: CDF/inverse CDF; non-central $\chi^2(\lambda)$. **Bias estimation:** fit model p(x,y), compute true I, draw n samples M times, compare $\hat{I}$ to I.

L8 · Information geometry

- **Euclid:** $s^2=z^TGz$, G=I. Manifold: locally like R^n; $ds^2=dx^Tg(x)dx$. Polar: $g=\text{diag}(1,r^2)$. Statistical manifold: points=PDFs; coordinates matter (σ^2 vs precision $\Lambda=\Sigma^{-1}$).
- **Divergence** D:M×M→[0,∞): D(p,q)≥0; =0 iff p=q; D(p,p+dp) positive-definite quadratic form. KL satisfies (Gibbs). Others: Bregman, f-divergences.
- **Projection:** idempotent P=P·P; Euclid: $\text{argmin}\|v-w\|^2$. **I-projection** $q_I^*=\text{argmin}_{q\in Q}D(q\|p)$: **zero-forcing** (q=0 where p=0), expectation wrt q, approximates complex with simple. **M-projection** $q_M^*=\text{argmin}\, D(p\|q)$: **zero-avoiding** (q>0 where p>0), expectation wrt p ⇒ usable with data samples. Differ since KL asymmetric.
- **MLE = M-projection:** $D(p\|q)=-H(p)-E_p[\log q]\approx -H(p)-(1/n)\Sigma\log q(x_i)$; argmax $E_p\log q=\text{argmin}\, D(p\|q)$.
- **MI = M-projection onto factorisable F:** $I(X;Y)=\min_{q\in F}D(p\|q)$ (=−H(X,Y)+H(X)+H(Y); optimal f=p(x), g=p(y)). Cond.: $I(X;Y|Z)=\min_{q\in C}D(p\|q)$, C={q:q(x|y,z)=q(x|z)}.
- **Pythagoras:** r=M-projection of p on (dually flat) Q, q∈Q: **$D(p\|q)=D(p\|r)+D(r\|q)$** (also for I-proj). With F⊂C: $I(X;Y,Z)=I(X;Y|Z)+I(X;Z)\Rightarrow$ MI chain rule.
- **Max entropy (Jaynes):** among models with $E[f_i(X)]=\alpha_i$, choose max H. Result: **$p(x)=Z^{-1}\exp(-\Sigma\lambda_if_i(x))$** (exponential family; Lagrangian: $-1-\log p+\Sigma\lambda_if_i=0$), λ set to meet constraints. Maxent = I-projection of uniform u onto Q={E_q $f_i=\alpha_i$} ($D(q\|u)=-H(q)+\text{const}$). More constraints ⇒ $H(q_2^*)\le H(q_1^*)$.
- **Fisher information:** $D(\theta\|\theta+d\theta)=½d\theta^TI(\theta)d\theta+O(d\theta^3)$ (offset & 1st derivative vanish). **$I(\theta)=-E[\partial^2\log p_\theta(X)/\partial\theta^2]$** (Hessian $I_{jk}=-E[\partial^2\log p/\partial\theta_j\partial\theta_k]$) = natural metric; = peak sharpness of likelihood; = **Var(score)**, score=$\partial\log p_\theta/\partial\theta$; = $E[\partial_i\log p\,\partial_j\log p]$ (first derivs only, autodiff). Gaussian (μ,σ): $I=\text{diag}(\sigma^{-2},2\sigma^{-2})$.
- **Bias-variance:** $MSE(\hat{\theta})=E[(\theta-\hat{\theta})^2]=$**Bias²+Var** (cross term=0 as θ fixed). **Cramér-Rao:** $MSE\ge b(\theta)^2+(1+b'(\theta))^2/I(\theta)$; unbiased: $Var\ge 1/I(\theta)$. Origin of trade-off is geometric.
- **Quick-reference:** I-proj ⇒ KL(q‖p), q forced to 0 where p=0 (mode-seeking); M-proj ⇒ KL(p‖q), covers all p support (mass-covering), = MLE. Projecting Gaussian p onto diagonal-Σ Gaussians ($Z_1\perp Z_2$): M-projection matches marginals, I-projection is narrower. Fisher info large ⇒ sharp likelihood ⇒ low variance bound; KL curvature = I.
- **Unification:** same formulae from communication (compression & transmission), hypothesis tests (LRT) and projections on statistical manifolds.